

XAI Agent Analysis — CREDIT DOMAIN

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RESPONSIBLE AI SCORE		
93 / 100		A ■ EXCELLENT
Explainability	Fairness	Compliance
30/30	36/40	27/30

Model Type	Domain	Accuracy	Test Samples
XGBoost	CREDIT	79.5%	200

EXECUTIVE SUMMARY

This Responsible AI Audit was conducted using the XAI Agent system, which combines SHAP explainability, LIME verification, fairness analysis using Fairlearn and IBM AIF360, and GDPR/EU AI Act compliance checking.

The model achieved an overall **Responsible AI Score of 93/100** with a status of **READY FOR DEPLOYMENT** . The EU AI Act classifies this system as **HIGH RISK**, requiring conformity assessment and human oversight before deployment.

Key findings include a fairness score of **60/100** indicating gender-based bias patterns in predictions, and a compliance score of **90/100** with most GDPR articles satisfied due to the explainability layer.

KEY FINDINGS

Finding	Status	Priority
SHAP + LIME explanations available	COMPLIANT	—
Borderline bias (sex_A92)	WARN	MEDIUM
Bias detected (sex_A93)	FAIL	CRITICAL
Bias detected (sex_A94)	FAIL	CRITICAL
Article 22	COMPLIANT	—
Article 5	COMPLIANT	—
Article 13	COMPLIANT	—
Article 25	COMPLIANT	—
Article 35	COMPLIANT	—
EU AI Act Risk Classification	HIGH RISK	CRITICAL

BIAS ANALYSIS

Overall Fairness Score: **60/100**

Protected Attribute	DPD	DPD Status	EOD	EOD Status
sex_A92	0.031746031746031744	PASS 	0.07164404223227752	WARN 
sex_A93	0.14170692431561993	FAIL 	0.13839285714285715	FAIL 
sex_A94	0.1888888888888888394	FAIL 	0.33199999999999996	FAIL 

COMPLIANCE ANALYSIS

Compliance Score: **90/100** | EU AI Act Risk: **HIGH**

Regulation	Severity	Status	Finding
Article 22	CRITICAL	COMPLIANT	SHAP and LIME explanations available
Article 5	HIGH	COMPLIANT	48 features is acceptable
Article 13	HIGH	COMPLIANT	Model decisions can be explained
Article 25	HIGH	COMPLIANT	Fairness score acceptable
Article 35	CRITICAL	COMPLIANT	DPIA completed and documented

REMEDIATION ROADMAP

Prioritized actions to improve the Responsible AI Score:

#	Priority	Action	Regulation
1	CRITICAL	Requires conformity assessment, human oversight, transp...	EU_AI_ACT
2	CRITICAL	Implement bias mitigation for sex_A93 and sex_A94	FAIRNESS
3	HIGH	Monitor proxy features: age, dependents correlation	FAIRNESS
4	MEDIUM	Add human review workflow for HIGH RISK decisions	EU_AI_ACT

This report was generated automatically by Rithvik's XAI Agent System. All findings should be reviewed by a qualified AI ethics professional before deployment.